

1D : $\xrightarrow{\hspace{2cm}}$ x ← koord. → tu se delec nahaja

klasično : $x(t) \rightarrow v(t) = \dot{x}(t)$
↙ namesto tira

kvantna : $\Psi(x,t)$ namesto Newtonov Zak.

$\Psi(x,0) \rightarrow i\hbar \frac{\partial \Psi(x,t)}{\partial t} = H \Psi(x,t) \rightarrow \Psi(x,t) \rightarrow \langle x,t \rangle = \int_{-\infty}^{\infty} \Psi^*(x,t) \times \Psi(x,t) dx$
↙ časovni razvoj naredimo s pomočjo Schrödingrove

$H = \frac{p^2}{2m} + V(x)$

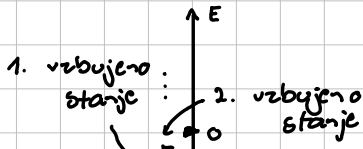
kje se v povprečju delec nahaja ob času t

$\langle p,t \rangle = \int_{-\infty}^{\infty} \Psi^*(x,t) p \Psi(x,t) dx$

tega se ponavadi ne piše $\rightarrow \hat{p} = -i\hbar \frac{\partial}{\partial x}$
 ↳ iz konteksta ugotovimo če je operator ali ne

Stacionarna Schrödingrova enačba :

$H \Psi_n^{(1)}(x) = E_n \Psi_n^{(2)}(x)$
↙ lastna vrednost → lastna energija
lastna funkcija

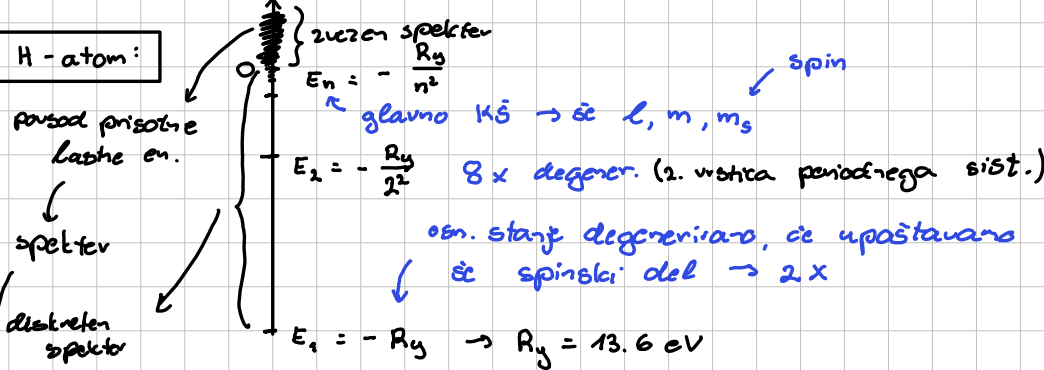


$H \Psi_n^{(2)}(x) = E_n \Psi_n^{(2)}(x)$

↳ imamo lahko 2 neodvisni l. fun. z isto lastno energijo → degenerirano stanje
 če ima energija samo eno stanje, je to stanje nede degenerirano.

osnomo stanje → E_1
 0 lahko izberemo → ne sme imeti opazljivih posledic

H-atom:



na nekem delu zvezan, na drugem diskreten → lahko imamo oboje pri istem primeru

Nede degenerirano stanje : $H \Psi(x) = E \Psi(x)$

$H[\lambda \Psi(x)] = E[\lambda \Psi(x)] \quad \lambda \in \mathbb{C}$

normalizacija $\int_{-\infty}^{\infty} |\Psi(x)|^2 dx = 1 \Rightarrow |\lambda| = 1, \lambda = e^{i\alpha}; \alpha \in \mathbb{R}$

$\Psi(x), e^{i\alpha} \Psi(x)$
 ↳ fazni faktor → to je še vedno normirana lastna fun.
 ↳ ne vpliva na nobeno opazljivo količino → svoboda matematike

$i\Psi(x) \rightarrow$ ni drugačija, fizika ostane enaka

degenerirano stanje : $H \Psi^{(1)} = E \Psi^{(1)}$

$\Psi^{(1)}, \Psi^{(2)}$ sta linearno neodvisni

npr. : 2 x $H \Psi^{(2)} = E \Psi^{(2)}$

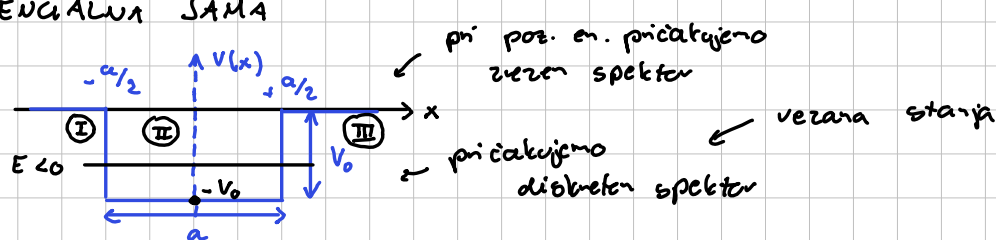
↓

$H[\alpha \Psi^{(1)}(x) + \beta \Psi^{(2)}(x)] = E[\alpha \Psi^{(1)}(x) + \beta \Psi^{(2)}(x)] ; \alpha, \beta \in \mathbb{C}$

$\Psi(x,0) = \sum_n c_n \Psi_n(x) ; c_n = \int \Psi_n^*(x) \Psi(x,0) dx$
↙ lastne fun., lastn. en. dobimo iz

$\Psi(x,t) = \sum_n c_n e^{-i \frac{E_n}{\hbar} t} \Psi_n(x)$ stacionarne Schrödingrove enačbe

POTENCIALNA JAMA



Vešana lastna stanja končne potencialne jame

$$H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} + V(x) \psi(x) = E \psi(x) \rightarrow \text{linearna dif. en. } (\psi(x) \text{ je povsod v lin. obliki})$$

→ 2 lin. neodv. rešitvi, kar je 2. reda
→ lahko tvorimo lin. kombinacije

$$\psi_I(x) = A e^{\kappa x} + B e^{-\kappa x}; \quad \kappa = \sqrt{\frac{-2mE}{\hbar^2}} \quad (\psi_I(x) = A' \sinh \kappa x + B' \cosh \kappa x)$$

$$\psi_{II}(x) = C e^{ikx} + D e^{-ikx}; \quad k = \sqrt{\frac{2m(E+V_0)}{\hbar^2}} \quad (\psi_{II}(x) = C' \sin kx + D' \cos kx)$$

$$\psi_{III}(x) = F e^{\kappa x} + G e^{-\kappa x} \rightarrow \text{To moramo poiskati}$$

→ zaradi normalizacije

robni pogoji:

$$\psi_I(-a/2) = \psi_{II}(-a/2) \quad \psi_{II}(a/2) = \psi_{III}(a/2)$$

$$\psi_I'(-a/2) = \psi_{II}'(-a/2) \quad \psi_{II}'(a/2) = \psi_{III}'(a/2)$$

Netrivialne rešitve: ko $\det = 0$

$$\rightarrow \left(\det \begin{pmatrix} \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix} \right) (E) = 0$$

Reševanje: $H\psi(x) = E\psi(x)$, $H = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x)$, predpostavimo $V(-x) = V(x)$ sod

$$x \mapsto -x: \quad \psi(-x) \rightarrow \text{l. sim. ga}$$

$$H\psi(-x) = E\psi(-x) \quad \text{isto lastno energijo}$$

E ni degeneriran

$$\psi(-x) = e^{i\alpha} \psi(x), \quad \alpha \in \mathbb{R}$$

sta lin. odvisna

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Limitni primer:

1.) $u_0 \rightarrow \infty \rightarrow$ modra črta gre modro gor, presečišča pri polih

$$u = n\pi, \quad n \in \mathbb{N}$$

$$\left(\frac{n\pi}{a}\right)^2 = \frac{2m(E+V_0)}{\hbar^2} \rightarrow E = \frac{\hbar^2}{2m} \left(\frac{n\pi}{a}\right)^2 - V_0$$

2.) $V_0 \rightarrow \infty, \quad a \rightarrow 0 \rightarrow$ modra črta gre modro gor, presečišča pri polih

$$V(x) = -\lambda \delta(x)$$

$u_0 \rightarrow 0$ (ker imamo a^2)

→ takoj po izhodšču pade na 0 → ali imamo presečišče?

$$t_{\frac{u}{2}} = \sqrt{\left(\frac{u_0}{u}\right)^2 - 1} \rightarrow \text{naslednik: } u = u_0 - \epsilon$$

$$t_{\frac{u_0 - \epsilon}{2}} = \sqrt{\left(\frac{u_0}{u_0 - \epsilon}\right)^2 - 1} = \sqrt{\left(\frac{1}{1 - \frac{\epsilon}{u_0}}\right)^2 - 1}$$

$$\frac{u_0 - \epsilon}{2} = \sqrt{\left(1 + \frac{\epsilon}{u_0}\right)^2 - 1} = \sqrt{1 + 2\frac{\epsilon}{u_0} - 1} = \sqrt{2\frac{\epsilon}{u_0}}$$

$$\frac{u_0 - \epsilon}{2} = \sqrt{2\frac{\epsilon}{u_0}} \rightarrow u_0^3 - \frac{1}{4}u_0^4 + \left(\frac{u_0}{2}\right)^6 = a^2 \frac{2m}{\hbar^2} (E+V_0)$$

$$\frac{2mV_0}{\hbar^2} a^2 - \frac{1}{4} \left(\frac{2mV_0}{\hbar^2} a^2\right)^2 = a^2 \frac{2m}{\hbar^2} E + a^2 \frac{2m}{\hbar^2} V_0$$

$$E = -\frac{1}{4} \frac{2ma^2}{\hbar^2} V_0^2$$

$$E_0 = -\frac{m}{2\hbar^2} \lambda^2$$

edino vezano stanje

→ kako izgleda valovna funkcija?

ni zvezno odv. v 0, ker je potencial dovolj grob

$V(x) = -\lambda \delta(x) \rightarrow$ ta potencial smo obravnavali kot potencial končne potencialne jame

$$\psi(x) = A e^{-K_0|x|} \rightarrow \text{določiti moramo } A \text{ in } K_0$$

$$K_0 = \sqrt{\frac{2m}{\hbar^2} \frac{m}{2\hbar^2} \lambda^2} = \frac{m}{\hbar^2} \lambda, \quad A \rightarrow \text{normalizacija}$$

$$\int_{-\infty}^{\infty} |\psi|^2 dx = 2 \int_{-\infty}^{\infty} A^2 e^{-2K_0|x|} dx = 2A^2 \left(\frac{1}{-2K_0} e^{-2K_0|x|} \right) \Big|_{-\infty}^{\infty} = -\frac{1}{K_0} A^2 (0 - 1) \Rightarrow A^2 = K_0$$

če imamo eno samo amplitudo

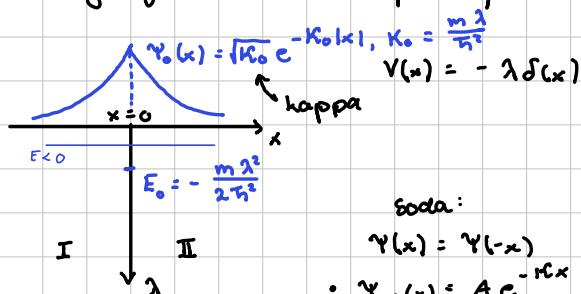
je lahko izbrano kot realno

(pogosto na 1. strani)

$$\psi(x) = \sqrt{K_0} e^{-K_0|x|}$$

→ to bomo pogosto uporabljali

nadaljevanje $\rightarrow$ želimo priti po lažji poti



soda:

$$\psi(x) = \psi(-x)$$

- $\psi_{II}(x) = A e^{-Kx}$
- $\psi_I(x) = A e^{Kx}$

$$H\psi(x) = E\psi(x)$$

$$H = \frac{p^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - \lambda \delta(x), \quad V(x) = V(-x)$$

lina:

$$\psi(x) = -\psi(-x)$$

- $\psi_{II}(x) = A e^{-Kx}$
- $\psi_I(x) = -A e^{-Kx}$

splošno:

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) - \psi(x) \lambda \delta(x) = E \psi(x) \quad \left/ \int_{0^-}^{0^+} dx \right. \quad \leftarrow \text{integral po intervalu katerega} \\ \text{sinja limitira proti 0}$$

$\leftarrow$ točka levo od 0, ampak limitno slizu

$$-\frac{\hbar^2}{2m} \frac{d}{dx} \psi(x) \Big|_{0^-}^{0^+} - \psi(0) \lambda = E \int_{0^-}^{0^+} \psi(x) dx$$

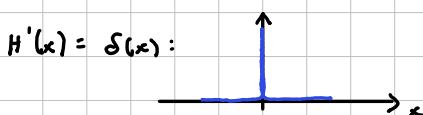
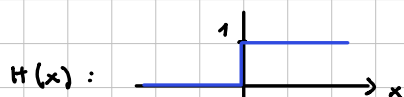
$\leftarrow$ to lahko preverimo na koncu

desni odvod $\leftarrow$ levi odvod

$$-\frac{\hbar^2}{2m} [\psi'(0^+) - \psi'(0^-)] - \psi(0) \lambda = 0$$

$= 0$, ker je $\psi(x)$ končna funkcija (integ. kon. fun. po so majhnem intervalu)

$$\psi'(0^+) - \psi'(0^-) = -\psi(0) \lambda \frac{2m}{\hbar^2} = -\psi(0) 2K_0 \neq 0 \rightarrow \text{levi in desni odvod nista enaka} \\ \rightarrow \text{fun. ne bo zvezno odvedljiva}$$



Če $\psi(x)$ nezvezna, dobimo $\delta'(x) \rightarrow$ se tuja singularnost od $\delta(x) \rightarrow$ ne moremo tega matematično rešiti $\Rightarrow \psi(x)$ mora biti zvezna $\rightarrow$ R.P.

$$\psi(0^-) = \psi(0^+) \rightarrow \text{zveznost}$$

$\rightarrow$ avtomatsko izpolnjen za soda stanja

2. A.P.: $\psi'(0^+) = -AK e^{-K0^+}$

$$\psi'(0^-) = AK e^{K0^-}$$

$$-AK - AK = -\psi(0) \cdot 2K_0 = -A \cdot 2K_0 \Rightarrow K = K_0$$

$$\psi(x) = A e^{-K_0|x|}$$

$$\psi'(0^+) - \psi'(0^-) = -2K_0 \psi(0)$$

$$\psi(0) = \psi(0^+) = \psi(0^-)$$

$\rightarrow$ robna pogoja za vedno ko imamo δ funkcijo, lahko je δ g. v kakšni drugi točki (ne v 0)

Se vedno nadaljeva:

$$\psi_1(x) = A_1 e^{ik_1 x} + B_2 e^{-ik_1 x}$$

$$k_1 = \sqrt{\frac{2m(E-V_1)}{\hbar^2}}$$

$$\psi_2(x) = B_1 e^{ik_2 x} + A_2 e^{-ik_2 x}$$

$$k_2 = \sqrt{\frac{2m(E-V_2)}{\hbar^2}}$$

dogovor: A-jji potujejo proti sipalcu, B-jji ven

Verjetnostni tok:
$$j(x) = \frac{\hbar}{2mi} \left[\underbrace{\psi^*(x) \psi'(x)}_z - \underbrace{\psi(x) \psi'^*(x)}_{z^*} \right] = \frac{\hbar}{m} \Im_m \{ \psi^*(x) \psi'(x) \}$$

$$j_1(x) = \frac{\hbar}{m} \Im_m \{ (A_1^* e^{-ik_1 x} + B_1^* e^{ik_1 x}) \cdot (A_1 k_1 i e^{ik_1 x} - B_1 k_1 i e^{-ik_1 x}) \} =$$

$$= \frac{\hbar}{m} \Im_m \{ |A_1|^2 k_1 i - |B_1|^2 k_1 i - \underbrace{A_1^* B_1 k_1 i e^{-2ik_1 x}}_z + \underbrace{A_1 B_1^* k_1 i e^{2ik_1 x}}_{z^*} \}$$

$z + z^* \in \mathbb{R}$, nas pa zanima samo $\Im_m$

$$j_1(x) = \frac{\hbar k_1}{m} (|A_1|^2 - |B_1|^2) \quad ; \quad j_2(x) = \frac{\hbar k_2}{m} (|B_2|^2 - |A_2|^2)$$

Spremenimo konstante (valove "normaliziramo"):

$$\psi_1(x) = \frac{A_1}{\sqrt{v_1}} e^{ik_1 x} + \frac{B_1}{\sqrt{v_1}} e^{-ik_1 x}$$

$$\psi_2(x) = B_2 \frac{e^{ik_2 x}}{\sqrt{v_2}} + A_2 \frac{e^{-ik_2 x}}{\sqrt{v_2}}$$

$$\Rightarrow j_1(x) = |A_1|^2 - |B_1|^2$$

$$j_2(x) = |B_2|^2 - |A_2|^2$$

Območje 3: velja Schrödinger $\rightarrow$ lin. dif. en. 2. reda $\rightarrow$ res.: lin. komb. 2 lin. neodv. res.
(to vedno lahko naredimo)

$$\psi_3(x) = C \varphi(x) + D \chi(x)$$

R.P.: (predpostavimo, da ni δ fun.)

- $\psi_1(a) = \psi_3(a)$
- $\psi_1'(a) = \psi_3'(a)$
- $\psi_2(b) = \psi_3(b)$
- $\psi_2'(b) = \psi_3'(b)$

$\rightarrow$ 4 enačbe, 6 amplitud:

to bi morali naprej poznati $\rightarrow$ mi pošljemo signal
4) nehomogen sistem z enolično rešitvijo

$$\rightarrow \text{lahko dobimo: } \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$\hookrightarrow$ sipalna matrika (S) $\rightarrow$ elementi kompleksni

$$\begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} r \\ t \end{pmatrix} \rightarrow \text{če pošljemo delce iz leve}$$

$$\begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} t' \\ r' \end{pmatrix} \rightarrow \text{če pošljemo delce iz desne}$$

$$j_1(x) = 1 - |r|^2$$

$$j_2(x) = |t|^2$$

$$1 = |r|^2 + |t|^2$$

$R = |r|^2$ $T = |t|^2$

Pri relativistični KM delci lahko izginejo $\rightarrow$ te enačbe ne veljajo več, mi se ukvarjamo z ne-relat.
(toliko delcev kot pošljemo noter, jih gre tudi ven)

Proti sipalcu: $|A_1|^2 + |A_2|^2$

Stran od sipalca: $|B_1|^2 + |B_2|^2$

morajo biti enaki $\rightarrow$ toliko kot gre noter, gre tudi ven; gledamo stacionarno stanje

by Dragicević

bodalo/dagger

$$|A_1|^2 + |A_2|^2 = \begin{pmatrix} A_1^* & A_2^* \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = A^H A = A^+ A$$

$$A = \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$$\Rightarrow A^+ B = B^+ B$$

S je unitarna matrika

$$\forall A: A^+ A = (SA)^+ SA = A^+ S^+ SA \Rightarrow S^+ S = I$$

$$\begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = \begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix}$$

$$B = SA, \quad S^+ S = I$$

spektralno to vedno velja

$$\psi^*(x) \rightarrow \psi_1^*(x) = A_1^* \frac{e^{-k_1 x i}}{\sqrt{v_1}} + B_1^* \frac{e^{ik_1 x}}{\sqrt{v_1}}$$

$$\psi_2^*(x) = B_2^* \frac{e^{-k_2 x i}}{\sqrt{v_2}} + A_2^* \frac{e^{ik_2 x}}{\sqrt{v_2}}$$

$$\begin{pmatrix} A_1^* \\ A_2^* \end{pmatrix} = S \begin{pmatrix} B_1^* \\ B_2^* \end{pmatrix}$$

$$A^* = S B^+ \rightarrow S^+ A^* = S^+ S B^+$$

$$S^+ A^* = B^+ / \rightarrow \boxed{S^+ A = B}$$

velja, če je Hamiltonian realen

$$S = \begin{pmatrix} r & t \\ t & r' \end{pmatrix}$$

S je simetrična $\leftarrow S^T = S$

H lahko postavimo $\in \mathbb{C}$ preko kin. en. (v magn. polju): $H = \frac{(\vec{p} - e\vec{A})^2}{2m} + V(\vec{r})$
 $\vec{p} = -i\hbar \nabla$

Simetrijska lastnost: $H\psi(x) = E\psi(x) \mid x \rightarrow -x, V(x) = V(-x) \mid k_1 = k_2 = k, v_1 = v_2 = v$

$$H\psi(-x) = E\psi(-x)$$

$$\psi(-x): \psi_1 = B_2 \frac{e^{-ikx}}{\sqrt{v}} + A_2 \frac{e^{ikx}}{\sqrt{v}}$$

$$\psi_2 = A_1 \frac{e^{-ikx}}{\sqrt{v}} + B_1 \frac{e^{ikx}}{\sqrt{v}}$$

$$\sigma_x^2 = I$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = S, \quad \begin{pmatrix} A_2 \\ A_1 \end{pmatrix} = S \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \Rightarrow \sigma_x B = S \sigma_x A \mid \sigma_x \text{ z leve}$$

$$\boxed{B = \sigma_x S \sigma_x A}$$

$$V(x) = V(-x) \Rightarrow S = \sigma_x S \sigma_x \mid S = \begin{pmatrix} r & t \\ t & r' \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} t & r' \\ r & t' \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} r' & t \\ t' & r \end{pmatrix}$$

$\rightarrow$ izračunati moramo z sipalno matriko
za protini potekalce varisan zgoraj velja

→ računalniška

$$\begin{pmatrix} r \\ t \end{pmatrix} = \begin{pmatrix} r & t' \\ t & r' \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$1 \cdot \frac{e^{ikx}}{\sqrt{v}} + r \frac{e^{-ikx}}{\sqrt{v}}$$

$\psi_1(x)$

$$k_1 = k_2 = k$$

$$v_1 = v_2 = v$$

$$V(x) = -\lambda \delta(x)$$

$$t \frac{e^{ikx}}{\sqrt{v}} = \psi_2(x)$$

$$\psi_1(0) = \psi_2(0)$$

$$K_0 = \frac{m\lambda}{\hbar^2}$$

$$[\psi_1'(0) - \psi_2'(0)] = -2K_0 \psi(0)$$

$$\frac{1}{\sqrt{v}} + \frac{r}{\sqrt{v}} = \frac{t}{\sqrt{v}}$$

$1+r=t$

$$\psi_1' = \frac{ik}{\sqrt{v}} e^{ikx} - \frac{r \cdot ik}{\sqrt{v}} e^{-ikx}$$

$$\psi_2' = \frac{tik}{\sqrt{v}} e^{ikx}$$

$$\left[\frac{tik}{\sqrt{v}} - \left\{ \frac{ik}{\sqrt{v}} (1-r) \right\} \right] = -2K_0 \frac{t}{\sqrt{v}}$$

$$\frac{ik}{\sqrt{v}} [t - 1 + r] = -2K_0 \frac{t}{\sqrt{v}}$$

2 enačbi

$$ik[1+r-1+r] = -2K_0(1+r)$$

$$ik2r = -2K_0(1+r)$$

$$ikr = -K_0 - K_0 r \Rightarrow r = \frac{-K_0}{ik + K_0}$$

$$t = 1 + r = \frac{ik + K_0 - K_0}{ik + K_0} = \frac{ik}{ik + K_0}$$

$$S = \frac{1}{ik + K_0} \begin{bmatrix} -K_0 & ik \\ ik & -K_0 \end{bmatrix}$$

$$T(E) = |t|^2 = \left| \frac{ik}{ik + K_0} \right|^2 = \frac{k^2}{k^2 + K_0^2}$$

$$= \frac{E}{E + \underbrace{\frac{\hbar^2 K_0^2}{2m}}_{E_0}}$$

$$E_0 \equiv \frac{\hbar^2 K_0^2}{2m} = \frac{m\lambda^2}{2\hbar^2}$$

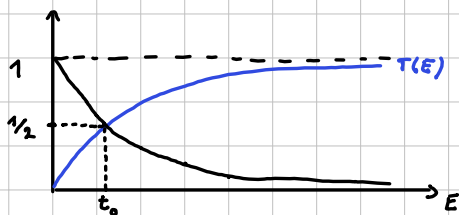
$$R(E) = \left| \frac{-K_0}{ik + K_0} \right|^2 = \frac{K_0^2}{k^2 + K_0^2} = \frac{E_0}{E + E_0}$$

$$T(E) + R(E) = 1$$

$$K_0 = \frac{m\lambda}{\hbar^2}$$

$$k = \sqrt{\frac{2m(E-0)}{\hbar^2}}$$

$$\Rightarrow E = \frac{\hbar^2 k^2}{2m}$$



$T(E) \rightarrow$ verjetnost za prepustnost
 $R(E) \rightarrow$ verjetnost za odbojnost

s kakšno cr. jr delac
 v zar, ko je v vztrajnem stanju
 to smo izpeljali pred
 2 tednoma

Novo !

$$\delta x \cdot \delta p \geq \frac{\hbar}{2} \rightarrow \text{želimo } \psi, \text{ da bo } \delta x \cdot \delta p = \frac{\hbar}{2}$$

V KM se sebi-adj. operatorji reče hermitski

$$\delta A \cdot \delta B \quad \begin{matrix} A^\dagger = A \\ B^\dagger = B \end{matrix} \rightarrow \text{Predpostavimo, da stvar ni problematična.}$$

$$\langle A \rangle \in \mathbb{R}, \langle B \rangle \in \mathbb{R}$$

$$A\psi = \psi'$$

$$\langle A \rangle \psi = \psi''$$

$$\tilde{A} \psi = \psi'''$$

$$\tilde{A} \equiv A - \langle A \rangle$$

$$\tilde{A}^\dagger = \tilde{A}$$

$$\hookrightarrow \langle A \rangle^\dagger = \langle A \rangle, \text{ ker je } \langle A \rangle \in \mathbb{R}$$

$$\tilde{B}^\dagger = \tilde{B}$$

$$\delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2} = \sqrt{\langle (A - \langle A \rangle)^2 \rangle}$$

$$\delta^2 A = \langle (A - \langle A \rangle)^2 \rangle \rightarrow \delta^2 x \cdot \delta^2 p \geq \left(\frac{\hbar}{2}\right)^2$$

$$\delta^2 A \cdot \delta^2 B = \langle \tilde{A}^2 \rangle \langle \tilde{B}^2 \rangle$$

Ponovitev Diracovega zapisa:

$$\langle C \rangle = \int dx \psi^*(x) C \psi(x) = \langle \psi | C | \psi \rangle = \langle \psi | C \psi \rangle = \langle C^\dagger \psi | \psi \rangle$$

sk. pr.



kvadrat dolžine vektorja

Cauchy-Schwarzova neenakost

$$\delta^2 A \cdot \delta^2 B = \langle \psi | \tilde{A}^2 | \psi \rangle \langle \psi | \tilde{B}^2 | \psi \rangle = \langle \tilde{A} \psi | \tilde{A} \psi \rangle \langle \tilde{B} \psi | \tilde{B} \psi \rangle \geq |\langle \tilde{A} \psi | \tilde{B} \psi \rangle|^2 =$$

$$\langle \psi | \tilde{A}^2 \psi \rangle = \langle \psi | \tilde{A} \tilde{A} \psi \rangle = \langle \tilde{A} \psi | \tilde{A} \psi \rangle$$

$$= |\langle \psi | \left(\frac{\{\tilde{A}, \tilde{B}\}}{2} + \frac{[\tilde{A}, \tilde{B}]}{2} \right) \psi \rangle|^2$$

vsota kvadratov

$$\tilde{A}\tilde{B} = \frac{\tilde{A}\tilde{B} + \tilde{B}\tilde{A}}{2} + \frac{\tilde{A}\tilde{B} - \tilde{B}\tilde{A}}{2} \rightarrow \frac{\{\tilde{A}, \tilde{B}\}}{2} + \frac{[\tilde{A}, \tilde{B}]}{2}$$

antikomutator

komutator

lastnosti komutatorjev in antikomutatorjev:

$$[\tilde{A}, \tilde{B}]^\dagger = (\tilde{A}\tilde{B} - \tilde{B}\tilde{A})^\dagger = (\tilde{A}\tilde{B})^\dagger - (\tilde{B}\tilde{A})^\dagger = \tilde{B}^\dagger \tilde{A}^\dagger - \tilde{A}^\dagger \tilde{B}^\dagger = \tilde{B}\tilde{A} - \tilde{A}\tilde{B} = [\tilde{B}, \tilde{A}] = -[\tilde{A}, \tilde{B}]$$

$$\{\tilde{A}, \tilde{B}\}^\dagger = (\tilde{A}\tilde{B} + \tilde{B}\tilde{A})^\dagger = (\tilde{A}\tilde{B})^\dagger + (\tilde{B}\tilde{A})^\dagger = \tilde{B}^\dagger \tilde{A}^\dagger + \tilde{A}^\dagger \tilde{B}^\dagger = \tilde{B}\tilde{A} + \tilde{A}\tilde{B} = \{\tilde{A}, \tilde{B}\}$$

$$C^\dagger = -C \rightarrow \text{antihermitski operator}$$

$$C^\dagger = C \rightarrow \text{hermitski op.}$$

$$\langle C \rangle = \langle \psi | C \psi \rangle = \langle -C \psi | \psi \rangle = \langle \psi | -C \psi \rangle^* = -C^*$$

$$\langle C \rangle = \langle \psi | C \psi \rangle = \langle C^\dagger \psi | \psi \rangle = \langle \psi | C \psi \rangle^* = C^*$$

$$\hookrightarrow \langle C \rangle \in \mathbb{R}$$

$$\langle C \rangle \in i\mathbb{R} \in \mathbb{R}$$

$$= \langle \psi | \frac{\{\tilde{A}, \tilde{B}\}}{2} \psi \rangle^2 + |\langle \psi | \frac{[\tilde{A}, \tilde{B}]}{2} \psi \rangle|^2$$

$$\geq |\langle \psi | \frac{[\tilde{A}, \tilde{B}]}{2} \psi \rangle|^2 \quad \text{če razpišemo, se vsi členi, kjer sta } \langle A \rangle \text{ in } \langle B \rangle, \text{ pokrajšajo}$$

$$[\tilde{A}, \tilde{B}] = [A - \langle A \rangle, B - \langle B \rangle] = [A, B]$$

$$[C, D] = 0, CD - DC = 0 \rightarrow CD = DC$$

$$= |\langle \psi | \frac{[A, B]}{2} \psi \rangle|^2$$

$$\delta A \cdot \delta B \geq |\langle \frac{[A, B]}{2} \rangle| \quad \delta x \cdot \delta p \geq |\langle \frac{[x, p]}{2} \rangle|$$

$$[x, p] \psi(x) = (xp - px) \psi(x) = \left(-x i \hbar \frac{d}{dx} - (i \hbar \frac{d}{dx}) x \right) \psi(x)$$

$$\hookrightarrow -i \hbar \frac{d}{dx}$$

$$= -i \hbar x \psi'(x) + i \hbar (1 \cdot \psi(x) + x \psi'(x)) = i \hbar \psi(x) \Rightarrow [x, p] = i \hbar$$

$$\delta x \cdot \delta p \geq \frac{\hbar}{2}$$

→ želimo enakost

1.) Pogoji pri C-S neenakosti: vektorja sta vzporedna

$$\tilde{B} |\psi\rangle = \lambda \tilde{A} |\psi\rangle$$

$$2.) \text{ pogoj pri drugem } \geq : \langle \psi | \{A, B\} \psi \rangle^2 = 0 = \langle \psi | (\tilde{A}\tilde{B} + \tilde{B}\tilde{A}) \psi \rangle =$$

$$= \langle \psi | \tilde{A}\tilde{B} \psi \rangle + \langle \psi | \tilde{B}\tilde{A} \psi \rangle = \langle \tilde{A} \psi | \tilde{B} \psi \rangle + \langle \tilde{B} \psi | \tilde{A} \psi \rangle = \langle \tilde{A} \psi | \lambda \tilde{A} \psi \rangle + \langle \lambda \tilde{A} \psi | \tilde{A} \psi \rangle =$$

$$= \lambda \langle \tilde{A} \psi | \tilde{A} \psi \rangle + \lambda^* \langle \tilde{A} \psi | \tilde{A} \psi \rangle = (\lambda + \lambda^*) \langle \tilde{A} \psi | \tilde{A} \psi \rangle$$

→ če izpostavimo iz prvega faktorja, ga moramo konjugirati

$\lambda \in i\mathbb{R}$ ali $\langle \tilde{A} \psi | \tilde{A} \psi \rangle = \delta^2 A = 0$ tedaj $\delta B \rightarrow \infty$ to nas ne zanima

$$\lambda \in i\mathbb{R}; \lambda \equiv i\mu, \mu \in \mathbb{R}$$

$$\tilde{B} |\psi\rangle = i\mu \tilde{A} |\psi\rangle$$

$$\tilde{p} |\psi\rangle = i\mu \tilde{x} |\psi\rangle \rightarrow (p - \langle p \rangle) |\psi\rangle = i\mu (x - \langle x \rangle) |\psi\rangle$$

$$|\psi\rangle \rightarrow \psi(x)$$

$$(-i \hbar \frac{d}{dx} - \langle p \rangle) \psi(x) = i\mu (x - \langle x \rangle) \psi(x)$$

$$-i \hbar \frac{d\psi(x)}{dx} = (i\mu (x - \langle x \rangle) + \langle p \rangle) \psi(x)$$

$$\int \frac{d\psi}{\psi} = \int \frac{1}{-i \hbar} (i\mu (x - \langle x \rangle) + \langle p \rangle) dx$$

$$\ln \psi = \frac{1}{-i \hbar} \left(i\mu \langle x \rangle x + \langle p \rangle x - \mu i \cdot \frac{x^2}{2} \right) + C'$$

$$\psi(x) = C e^{-\frac{\mu}{\hbar} x \langle x \rangle - \frac{\langle p \rangle x}{i \hbar} + \frac{\mu}{\hbar} \frac{x^2}{2}}$$